

Flight Operational Contact Plan

Satellite Flight Operator Technical Assessment - Part 1

Prepared by:

ANASTASIA MARIA GIANNAKOPOULOU

August 2026

1. Purpose

This document defines the procedure for conducting the assigned 10-minute ground-station contact, including spacecraft health verification, data transfer and the Communications Subsystem configuration change.

1.1 Operational Objectives

The procedure is designed to:

- ✓ Establish and maintain situational awareness of spacecraft status.
- ✓ Execute mission activities within the available contact time.
- ✓ Apply the Communications Subsystem configuration only after required data transfers are complete.
- ✓ Return the spacecraft to a verified communications state before LOS.

2. Contact Parameters

The following operational parameters are defined by the assessment and form the baseline constraints for the contact procedure. Additional assessment-provided requirements, including the required activities and Communications Subsystem behaviour, are addressed in the subsequent sections.

Table 1 - Assessment-Defined Contact Parameters

PARAMETER	VALUE
Contact Duration	10 minutes
Uplink Rate	50 kbps
Downlink Rate	50 kbps
RX	Permanently ON
TX	Commanded ON when required
AOCS	Fully automated

The communications link is assumed to operate in full-duplex mode, with RX permanently ON and TX commanded ON when required.

The Communications Subsystem configuration change requires a subsystem restart. Following the restart, TX Modulation must be commanded again to restore the downlink.

3. Activity Dependencies and Risks

The activities are sequenced according to their operational dependencies.

Table 2 - Activity Dependencies

ACTIVITY	PREREQUISITE	CONSEQUENCE IF UNAVAILABLE
Spacecraft health check	Valid telemetry	Spacecraft state cannot be confirmed
Payload data transfer	Health check passed; downlink available	Payload data remains undelivered
Platform data transfer	Health check passed; downlink available	Platform data remains undelivered
Configuration upload	Uplink available; valid configuration file	Configuration change cannot proceed
COMMS restart	Configuration successfully uploaded	New configuration cannot be applied
TX Modulation restoration	COMMS restart completed	Downlink remains unavailable
Final verification	Downlink restored; valid telemetry	Final spacecraft state cannot be confirmed

3.1. Risk Assessment

The following risks are considered when planning the contact.

Table 3 - Operational Risk Assessment

RISK	IMPACT	MITIGATION
Loss of downlink	Telemetry and data transfer unavailable	Prioritise critical activities and retain sufficient time for recovery
Insufficient contact time	Planned activities may remain incomplete	Prioritise data and configuration activities before LOS
Invalid (or incorrect) configuration	Communications may not operate as expected after restart	Verify configuration file and upload status before restart
COMMS restart does not complete as expected	Downlink may not be restored	Follow the applicable recovery procedure and avoid (unnecessary) further commanding
TX Modulation not restored after restart	Spacecraft remains without downlink	Treat TX Modulation restoration as a mandatory post-restart activity
Unexpected spacecraft telemetry	Potential spacecraft or subsystem fault	Stop non-essential commanding and assess telemetry before proceeding

4. Data Transfer Priority and Contact-Time Allocation

4.1 Data Transfer Priority

Payload data is assigned as the highest data-transfer priority because it represents the primary mission output. Platform data is assigned as the second priority and is transferred when the remaining contact time permits.

Table 4 - Data Transfer Priority

PRIORITY	DATA	RATIONALE
1	Payload data	Primary mission data and highest operational value
2	Platform data	Required for platform monitoring and analysis

4.2 Working Assumptions

The assessment does not specify the volume of payload or platform data to be transferred. The following values are therefore used as working assumptions for the contact-time analysis.

Table 5 - Working Data-Volume Assumptions

PARAMETER	ASSUMED VALUE	BASIS
Payload data	1.5 MB	Representative mission-data volume for this task
Platform data	0.5 MB	Representative platform-data volume for this task
Effective downlink rate	50 kbps	Assessment-defined

4.3 Transmission-Time Calculation

At the assessment-defined downlink rate of 50 kbps, the estimated transmission time is calculated as:

$$\text{TRANSMISSION TIME} = \text{DATA VOLUME} \times 8 / \text{DOWNLINK RATE}$$

Table 6 - Estimated Data-Transfer Times

DATA	VOLUME	ESTIMATED TRANSMISSION TIME
Payload	1.5 MB	240 s (4 min)
Platform	0.5 MB	80 s (1 min 20 s)
Total	2.0 MB	320 s (5 min 20 s)

Assuming decimal MB:

- 1.5 MB = 12 Mbit → 12,000 kb / 50 kbps = **240 s**
- 0.5 MB = 4 Mbit → 4,000 kb / 50 kbps = **80 s**

4.4 Contact-Time Allocation

The 10-minute contact is allocated between mandatory operational activities and a contingency margin. The allocation prioritises mission data transfer while reserving sufficient time for the Communications Subsystem configuration change, restart, TX Modulation restoration and final verification.

Table 7 - Proposed Contact-Time Allocation

Activity	Allocated time
Initial health check	60 s
Payload data transfer	240 s
Platform data transfer	80 s
Configuration upload	30 s
COMMS restart and recovery	90 s
TX Modulation Restoration	40 s
Final verification	40 s
Operational contingency margin	20 s
Total	600 s (10 min)

Note: Configuration upload, COMMS restart and recovery, TX Modulation restoration and final verification durations are planning assumptions used to demonstrate contact feasibility. Actual execution times would be replaced by validated spacecraft procedures and command execution times. The 20-second contingency margin is reserved for minor delays, command retries or additional verification and is not assigned to a nominal activity.

5. Operational Sequence

The nominal contact sequence below applies the activity dependencies and contact-time allocation defined in Sections 3 and 4. Each step requires confirmation of the expected spacecraft response before proceeding to the next dependent activity.

The Communications Subsystem configuration change is intentionally performed after the planned data transfers. The configuration application requires a subsystem restart, during which the downlink is lost. The restart is therefore initiated only after the configuration upload has been successfully verified and sufficient contact time remains for TX Modulation restoration and post-restart verification.

Table 8 - Nominal Contact Operational Sequence

STEP	TIME	OPERATOR ACTION	EXPECTED RESULT
1	T+00:00–01:00	Perform the initial spacecraft health check using available telemetry. Review spacecraft mode, power, thermal, AOCS and Communications Subsystem status.	Spacecraft health is confirmed within expected operational conditions.
2	T+01:00–05:00	Initiate payload-data download. Monitor downlink status and transfer progress throughout the activity.	Payload data is transferred successfully or the incomplete volume is recorded.
3	T+05:00–06:20	Initiate platform-data download. Continue monitoring the downlink and remaining contact time.	Platform data is transferred successfully or the incomplete volume is recorded.
4	T+06:20–06:50	Upload the approved Communications Subsystem configuration file and verify successful upload.	Configuration is confirmed available for application.
5	T+06:50–08:20	Command the Communications Subsystem restart and monitor recovery.	Communications Subsystem completes the restart and reaches the expected recovery state.
6	T+08:20–09:20	Command TX Modulation restoration and verify the return of the downlink.	Downlink is restored and valid telemetry is received.
7	T+09:20–10:00	Perform final spacecraft and Communications Subsystem verification. Record completed, incomplete or deferred activities and prepare for LOS.	Spacecraft state and completed/deferred activities are recorded; spacecraft is ready for LOS.

6. Contingency and Recovery Procedures

The contingency procedures define the response to conditions that could prevent the nominal contact sequence from being completed. They are included to limit the impact of anomalies on spacecraft safety, communications availability and mission data return.

The main mission impacts considered are:

- **Spacecraft safety:** An unexpected spacecraft state may require planned activities to be suspended.
- **Communications availability:** Loss or degradation of the downlink can prevent telemetry reception and data transfer.
- **Mission data return:** A failed or interrupted transfer may leave payload or platform data incomplete.
- **Configuration integrity:** An unsuccessful configuration upload or restart may prevent the intended Communications Subsystem configuration from being applied.
- **Contact-time utilisation:** An anomaly can reduce the time available for remaining activities and may require lower-priority operations to be deferred.
- **Post-contact state:** The spacecraft must be left in a known and documented state, including any incomplete or deferred activities.

The following contingencies define the operator response when an expected condition is not met. Non-essential activities are deferred when required to protect spacecraft safety, communications availability and sufficient time for recovery before LOS.

Table 9 - Contingency and Recovery Actions

CONDITION	OPERATOR RESPONSE	DECISION
Loss of downlink	Pause commanding and assess link/telemetry status.	Attempt approved recovery; otherwise terminate non-essential activities.
Unexpected telemetry	Pause the sequence and assess spacecraft state.	Resume only after the state is understood and safe.
Payload transfer failure	Verify link and transfer status.	Retry if time permits; otherwise defer remaining data.
Platform transfer failure	Verify link and transfer status.	Defer transfer if required to protect higher-priority activities.
Configuration upload failure	Verify upload status and file acknowledgement.	Do not initiate the COMMS restart; retry if permitted.
COMMS restart failure	Allow the defined recovery period and assess subsystem status.	Follow the approved recovery procedure; avoid repeated commands without confirmation.
TX Modulation not restored	Verify TX and Communications Subsystem status.	Retry only if authorised; otherwise end non-essential commanding.
Insufficient time before LOS	Review remaining activities and available time.	Defer any activity that cannot be safely completed before LOS.

7. End-of-Contact State

Before loss of signal (LOS), the operator shall verify the spacecraft state, confirm the status of completed and deferred activities, and ensure that any outstanding conditions are documented for the next contact.

Table 10 - Expected End-of-Contact State

PARAMETER	EXPECTED STATE AT LOS
Spacecraft health	Nominal, with no unresolved blocking anomaly
Communications Subsystem	Operational and verified
TX Modulation	Restored and verified after restart
Configuration	New configuration verified active, or previous configuration retained if the change was aborted
Payload data	Transferred successfully, or incomplete volume recorded for the next contact
Platform data	Transferred successfully, or incomplete volume recorded
Operational record	Commands, results, anomalies and deferred activities documented

8. Operational Decision Principles

The following principles guide operational decisions when the contact deviates from the nominal sequence:

- **Spacecraft safety:** A safe and understood spacecraft state takes priority over completing the contact plan.
- **Verify before proceeding:** Do not proceed with a dependent activity until the expected spacecraft response or state has been confirmed.
- **No “blind” commanding:** Do not issue repeated commands without understanding the spacecraft state and expected response.
- **Protect communications:** Do not initiate the Communications Subsystem restart unless the configuration upload has been verified and sufficient recovery time remains.
- **Protect contact time:** Continuously reassess the remaining contact time before retries or non-essential activities.
- **Data prioritisation:** For this task, payload data takes priority over platform-data transfer after the health check.
- **Known end state:** The spacecraft should reach LOS in a known and documented configuration/state.

End of Procedure

The contact is considered complete once the spacecraft state and operational results have been verified and documented, with any incomplete or deferred activities identified for subsequent operations.

— End of Flight Operational Contact Plan —